MD. NUROL AMIN

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ABOUT ME

I am an M.Sc. student in Computer Science and Engineering at Daffodil International University, working in quantum information science and its intersection with machine learning, network engineering and security. My B.Sc. thesis introduced Congestion-Triggered Distillation, a dual-threshold buffer-management policy for quantum-repeater entanglement memories; my M.Sc. thesis, NEMA-Q, develops a hyperbolic quantum-classical graph neural network with observable-level explainability, and the accompanying sole-author paper was accepted at ICEQT'26 for publication in Springer Nature proceedings. My current work spans quantum machine learning and explainability, QKD security, quantum error correction and networking, and quantum metrology and entanglement certification. Every project is simulation-first and reproducible from a fixed seed, and I report negative results — seed variance, minority-class collapse, robustness that traced back to a classical bypass — alongside the headline numbers.

EDUCATION

M. Sc. in Computer Science and Engineering (CSE) 2026 – Present
Daffodil International University, Dhaka
Thesis: *NEMA-Q — A Hyperbolic Quantum-Classical Graph Neural Network with Observable-Level Explainability*
Expected Graduation: May, 2027

B. Sc. in Computer Science and Engineering (CSE) 2022 – 2025
Daffodil International University, Dhaka
Thesis: *Congestion-Triggered Distillation — A Dual-Threshold Buffer-Management Policy for Quantum-Repeater Entanglement Memories*
CGPA: 3.45 (Out of 4.00)

Higher Secondary Certificate (HSC), Science 2018 – 2020
The Millennium Stars School And College, Rangpur
GPA: 5.00 (Out of 5.00)

Secondary School Certificate (SSC), Science 2016 – 2018
Dinajpur High School, Dinajpur
GPA: 5.00 (Out of 5.00)

ACHIEVEMENTS

Paper Accepted at ICEQT'26 / CSCE'26: Sole-author regular research paper accepted for Springer Nature proceedings (Scopus, EI, Web of Science indexed), Las Vegas, USA. 2026

Qiskit Global Summer School 2026: Completed the advanced lab track of IBM's flagship quantum computing summer program. 2026

Qiskit Global Summer School 2025 – Quantum Excellence: Awarded for outstanding performance in IBM's flagship quantum computing summer program. [Verify] 2025

IBM Quantum Machine Learning Certification: Certified in quantum machine learning through IBM Quantum Network. [Verify] 2025

QWorld Certifications: Completed QWorld quantum computing training and certification programs. 2024–2025

RESEARCH & PUBLICATIONS

Accepted

NEMA-Q: A Hyperbolic Quantum-Classical Graph Neural Network with Observable-Level Explainability, Accepted at ICEQT'26 – 5th International Conference on Emergent Quantum Technologies, part of CSCE'26 (Las Vegas, USA, July 20–23, 2026). Sole author. Proceedings published by Springer Nature; indexed in Scopus, EI Compendex, DBLP, ACM DL, and Clarivate Web of Science (CPCI).

Under Review

Which Quantum-Hacking Attacks Can a Decoy-State BB84 Receiver Detect Unaided?, Submitted to ICRPSET 2026 (IEEE track). Shows that QBER monitoring is structurally blind to photon-number-splitting and detector-blinding attacks, and that a compact classifier over observables a decoy-state receiver already computes detects all three attack families with no added hardware.

Manuscripts in Preparation

Entanglement-Assisted Covert Communication over Thermal-Loss Bosonic Channels: A Physically Consistent Framework with Finite-Blocklength and Collective-Adversary Analyses – Targeting IEEE Transactions on Quantum Engineering.

Congestion-Triggered Distillation: Turning Buffer Saturation into Entanglement Quality – Dual-threshold buffer-management policy for quantum-repeater entanglement memories. Targeting IEEE Transactions on Quantum Engineering.

Genuine Multipartite Entanglement Certification for a $2 \times 3 \times 3$ Composite-Dimension Qudit: Exact Thresholds and a Verified SDP Witness – Targeting IEEE Transactions on Quantum Engineering.

Adaptive Entanglement Allocation for Distributed Quantum Sensing under Drifting Local Noise – GHZ-block partitioning of a sensor array under heterogeneous, time-varying dephasing. Targeting Physical Review A / Quantum / Quantum Science and Technology.

Logical Teleportation between Heterogeneous QEC Codes over a Noisy Bell-Pair Interconnect – Surface code $\leftrightarrow$ bivariate-bicycle $[[72, 12, 6]]$ adapter with verified merged-code construction; link-infidelity thresholds and ebit budgets. Targeting PRX Quantum.

Magic-Anchored Initialization for Variational Quantum Circuits – Clifford/Clifford+T native parameter anchors with trainability certificates and classical-surrogate limits. Targeting PRX Quantum / Quantum.

Equivariant Post-Variational Quantum Machine Learning: Symmetry-Adapted Fixed Circuits with Convex Training Guarantees – Targeting Quantum.

Provably Trainable Lightweight QNNs: Dynamical-Lie-Algebra Dimension as a Barren-Plateau Criterion – Numerical verification that gradient variance scales with DLA dimension rather than Hilbert-space dimension.

Quantum Reservoir Computing for ECG Arrhythmia Classification with a Classical-Surrogate Advantage Audit – Targeting Physical Review Applied / Quantum / IEEE TQE.

Non-Markovian Memory Decay in Quantum-Repeater Rate Analysis: A Correction and Methods Paper – Shows the standard exponential-decay assumption systematically underestimates achievable entanglement rate for $1/f$-limited memories.

DATASETS & OPEN RESEARCH ARTIFACTS

QKD-IDS-v5: Physics-Grounded Synthetic Benchmark for Zero-Day Intrusion Detection on Decoy-State BB84
2026
52,000 labelled monitoring windows (40k normal / 12k attack) across six attack families, two held out for honest zero-day evaluation. Features grounded in decoy-state / GLLP theory with QBER recomputed from first principles per window. Released with a Datasheets-for-Datasets record, per-feature data dictionary, SHA-256 manifest, and a deterministic seed-42 regeneration path. Data CC-BY-4.0, code MIT.

RESEARCH EXPERIENCE

NEMA-Q: Hyperbolic Quantum-Classical Graph Neural Network 2026 – Present
Python, PennyLane, PyTorch, SHAP

- Built a hybrid quantum-classical GNN operating in hyperbolic (Poincaré) space for node classification, with a five-principle explainability suite.
- Introduced Quantum Observable Attribution (QOA): per-node gradients of class logits with respect to measured quantum observables, cross-checked against integrated gradients and SHAP over the fused representation.
- Ran component-isolating ablations over five seeds and reported negative results honestly, including seed variance, a minority-class collapse case, and evidence that noise stability originates in the classical bypass rather than intrinsic quantum robustness.
- Accepted as a regular research paper at ICEQT'26 (Springer Nature proceedings).

ML Detection of Quantum-Hacking Attacks on Decoy-State BB84 2026 – Present
Python, PyTorch, scikit-learn, NumPy

- Implemented a decoy-state BB84 simulator with physics-level intercept-resend, photon-number-splitting, and detector-blinding attack models, validated by ten simulator consistency checks.
- Benchmarked 1-D CNN, BiLSTM, and MLP detectors against two incumbent baselines (QBER threshold and decoy Y_1/e_1 bounds with gain-ratio consistency) at a matched false-alarm rate over five seeds.

- Achieved per-attack detection gains of +6.1, +14.8, and +66.6 points against the stronger incumbent for each attack, with online sliding-window latency measured at a 1% session false-alarm rate.
- Evaluated under distribution-shift and leave-one-attack-out protocols; released a tidy CSV export of the full dataset.

QKD-IDS: Deep SVDD Pipeline for Zero-Day QKD Intrusion Detection 2026 – Present
Python, PyTorch, scikit-learn

- Designed a 52,000-window physics-grounded benchmark for unsupervised intrusion detection on a 25 km decoy-state BB84 link, with two attack families held out entirely from training.
- Trained Deep SVDD one-class detectors and a fusion ensemble, evaluated leave-one-attack-out to measure genuine zero-day generalization rather than in-distribution recall.
- Packaged a fully deterministic release with checksum manifest, Croissant metadata, and a public leaderboard scaffold.

Cross-Code Logical Teleportation Simulator 2026 – Present
Python, Stim, NumPy, SciPy

- Constructed and verified a heterogeneous CSS adapter merging a distance-3 surface code with the bivariate-bicycle $[[72, 12, 6]]$ code, confirming commutation, logical count, and distance across five independent search seeds.
- Implemented a circuit-level Stim syndrome-extraction cycle for the BB code, deriving a valid six-layer CNOT schedule from bipartite edge-colouring of the provably 6-regular check graph.
- Diagnosed and fixed a determinism bug caused by mixing X- and Z-check CNOTs within a round, isolating the root cause through Heisenberg-picture analysis.

EA-CQC: Entanglement-Assisted Covert Quantum Communication Simulator 2025 – Present
Python, NumPy, SciPy, Matplotlib

- Implemented closed-form capacity analysis comparing unassisted Gaussian and entanglement-assisted (TMSV) schemes over thermal-loss bosonic channels.
- Computed quantum relative entropy, quantum Chernoff bound, finite-blocklength penalties, and Square Root Law scaling without approximation.
- Validated 21 analytical invariants including EA dominance, dispersion ordering, and monotone transmissivity-sweep behavior.

CTD: Quantum Repeater Buffer-Management Policy Simulator 2025 – Present
Python, NumPy, Matplotlib

- Designed and simulated a dual-threshold Congestion-Triggered Distillation policy for quantum-repeater entanglement memories under Poisson traffic.
- Benchmarked three-way policy comparison (Always-Distill, Never-Distill, CTD) across hardware-sensitivity and jitter-robustness scenarios.
- Demonstrated yield improvements and reduced memory-cutoff waste across parameter sweeps.

GME Certification Tool: SDP Witness for $2 \times 3 \times 3$ Qudit System 2025 – Present
Python, CVXPY, NumPy

- Computed exact fidelity-based thresholds for genuine multipartite entanglement in composite-dimension qudit systems via semidefinite programming.
- Derived and verified a PPT-criterion-based SDP witness with numerical certificate confirming GME above computed thresholds.
- Implemented decoherence model and hardware-noise robustness analysis for realistic near-term quantum devices.

SOFTWARE PROJECTS

Laptop IO — Android Phone as Wireless Keyboard and Trackpad 2026
Kotlin, Android SDK, Node.js, WebSockets

- Built and shipped a two-part product: a Kotlin Android client (trackpad, mouse buttons, keyboard, media keys) and a Node.js server that injects input events into Windows over the LAN.
- Secured the channel with TLS and a trust-on-first-use certificate pin, a fresh six-digit pairing code per launch, long-lived per-device tokens, schema validation on every inbound message, and per-IP brute-force lockout.

- Made power and open-URL capabilities opt-in by default, versioned the wire protocol, and matched the Windows Precision Touchpad acceleration curve so the phone pad feels native.
- Packaged with a signed release keystore, QR-code pairing, and one-click server launch.

SmartFarmX — Off-Grid Farm Monitoring System May 2025 – August 2025
IoT, LoRa, Web Development

- Developed a system to monitor remote farm fields without internet reliance using LoRa communication.
- Implemented real-time sensor data visualization and a responsive web interface for farm monitoring.

QKD-IDS-v5 — Public Benchmark Dataset Release 2026
Python, PyTorch, Croissant metadata

- Packaged a 52,000-window QKD intrusion-detection benchmark as a reusable public artifact: frozen CSV splits, per-feature data dictionary, Datasheets-for-Datasets record, SHA-256 manifest, and a leaderboard scaffold.
- Built a one-command deterministic regeneration path so any user can rebuild the release byte-for-byte from seed 42.

SKILLS

Programming Languages	Python, C, Java, HTML, CSS
Quantum Computing	Qiskit, PennyLane, Cirq, Stim, CVXPY (SDP), Quantum Circuit Design
Quantum Research Areas	QKD Security, Quantum Error Correction, Quantum Repeaters, Quantum Metrology, Entanglement Certification, Covert Communication
Machine Learning & AI	PyTorch, scikit-learn, Deep Learning, Graph Neural Networks, Computer Vision, Quantum Machine Learning, Explainable AI (SHAP, IG)
Research & Writing	LaTeX, Academic Writing, Research Methodology, Reproducible Pipelines
Tools & Platforms	Git, Google Colab, Kaggle, Jupyter, Microsoft Office

REFERENCES

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